ISTANBUL TECHNICAL UNIVERSITY

Department of Electrical Engineering

ENGINEERING DRAWING (ELK 363E)

Dimensioning

Assoc. Prof. Dr. Ali Sinan ÇABUK

ENGINEERING DESIGN

PROCESS	RESULT	TRANSFERRED INFORMATION
Design a part	Sketches of ideas	
Create drawings	Multiview Drawing	Shape
	Dimensioning	1. Size, Location 2. Non-graphic information
Manufacture		

TOPICS

- Introduction
- Dimensioning components
- Dimensioning object' s features
- Placement of dimensions.

DEFINITION

Dimensioning is the process of specifying part' s information by using of **figures, symbols** and **notes**.

This information are such as:

1. Sizes and locations of features
2. Material's type
3. Number required
This course
4. Kind of surface finish
5. Manufacturing process
6. Size and geometric tolerances

Introduction

DIMENSIONING SYSTEM

1. Metric system : ISO and JIS standards
Examples 32, 32.5, 32.55, 0.5 *(not .5)* etc.
This course

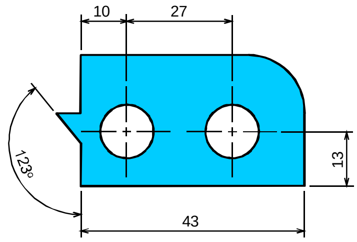
2. Decimal-inch system
Examples 0.25 *(not .25)*, 5.375 etc.

3. Fractional-inch system
Examples $\frac{1}{4}$, $5\frac{3}{8}$ etc.

Dimensioning Components

DIMENSION LINES

indicate the direction and extent of a dimension, and inscribe ***dimension figures***.



DIMENSIONING COMPONENTS

- Extension lines
- Dimension lines
  (with arrowheads)
- Leader lines

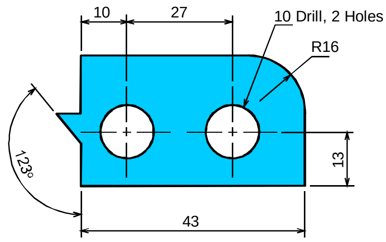
Drawn with
4H pencil

- Dimension figures
- Notes :
  - *local note*
  - *general note*

Lettered with
2H pencil.

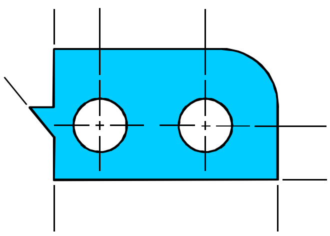
LEADER LINES

indicate details of the feature with a ***local*** note.



EXTENSION LINES

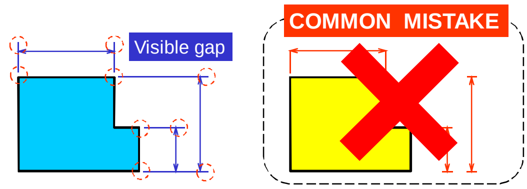
indicate the location on the object's features that are dimensioned.



Recommended Practices

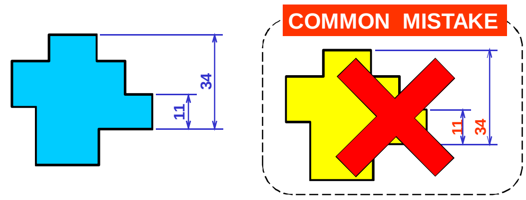
EXTENSION LINES

- Leave a **visible gap** (≈ 1 mm) from a view and start drawing an extension line.
- Extend the lines beyond the (last) dimension line 1-2 mm.



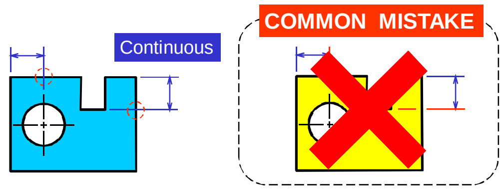
DIMENSION FIGURES

- The height of figures is suggested to be 2.5~3 mm.
- Place the numbers at about 1 mm *above dimension line* and *between extension lines*.



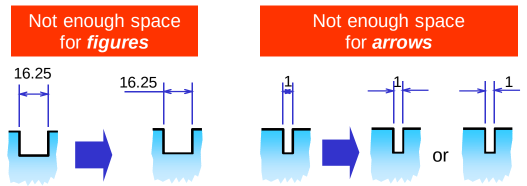
EXTENSION LINES

- **Do not** break the lines as they cross object lines.



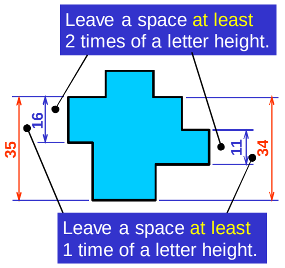
DIMENSION FIGURES

- When there is **not** enough space for figure or arrows, put it **outside** either of the extension lines.



DIMENSION LINES

- Dimension lines should **not** be spaced too close to each other and to the view.



DIMENSION FIGURES : UNITS

The JIS and ISO standards adopt the unit of

- **Length** dimension in **millimeters without** specifying a unit symbol “mm”.

- **Angular** dimension in **degree** with a symbol “o” place behind the figures (and if necessary **minutes** and **seconds** may be used together).

DIMENSION FIGURES : ORIENTATION

1. Aligned method

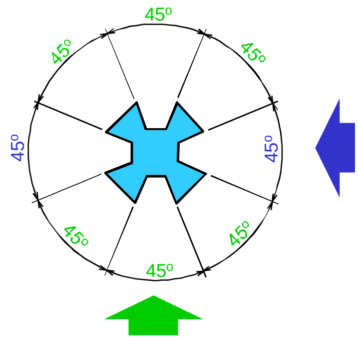
The dimension figures are placed so that they are readable from the **bottom** and **right side** of the drawing.

2. Unidirectional method

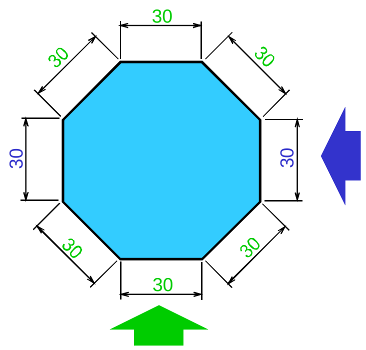
The dimension figures are placed so that they can be read from the **bottom** of the drawing.

Do not use both system on the same drawing or on the same series of drawing (JIS Z8317)

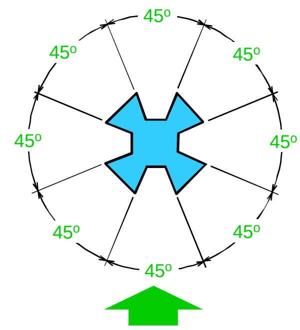
EXAMPLE : Dimension of ***angle*** using ***aligned*** method.



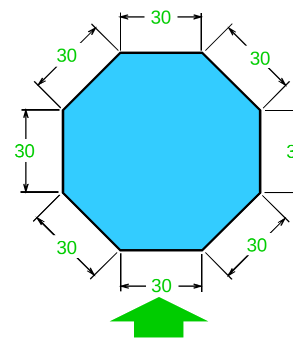
EXAMPLE : Dimension of ***length*** using ***aligned*** method.



EXAMPLE : Dimension of ***angle*** using ***unidirectional*** method.

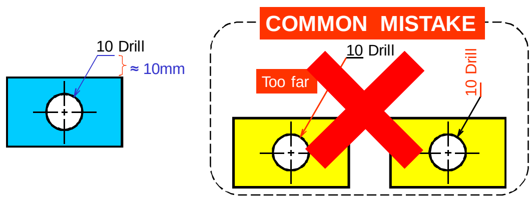


EXAMPLE : Dimension of ***length*** using ***unidirectional*** method.



LOCAL NOTES

- Place the notes **near** to the feature which they apply, and should be placed outside the view.
- Always read **horizontally**.



Dimensioning Practices

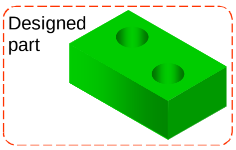
THE BASIC CONCEPT

Dimensioning is accomplished by adding ***size*** and ***location*** information ***necessary to manufacture*** the object.

This information have to be

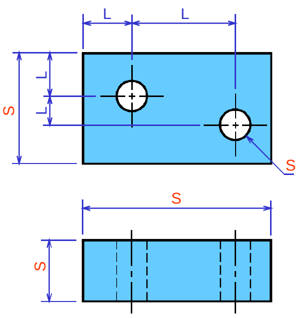
- Clear
- Complete
- Facilitate the
  - manufacturing method
  - measurement method

EXAMPLE



To manufacture this part we need to know…

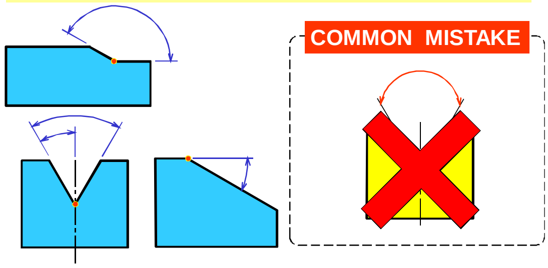
1. **Width**, **depth** and **thickness** of the part.
2. **Diameter** and **depth** of the hole.
3. **Location** of the holes.



"S" denotes size dimension.
"L" denotes location dimension.

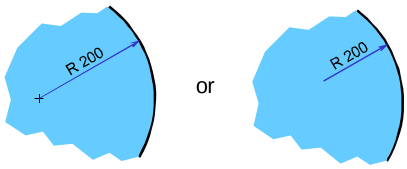
ANGLE

- To dimension an angle use **circular dimension line** having the center at the vertex of the angle.



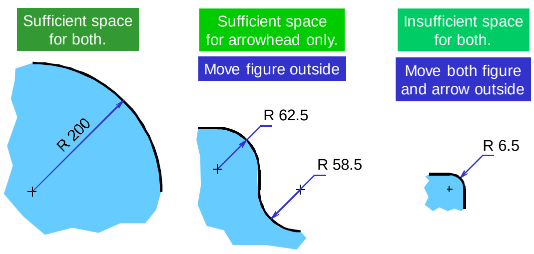
ARC

- Arcs are dimensioned by giving the **radius**, in the views in which their true shapes appear.
- The letter "R" is **always** lettered before the figures to emphasize that this dimension is radius of an arc.



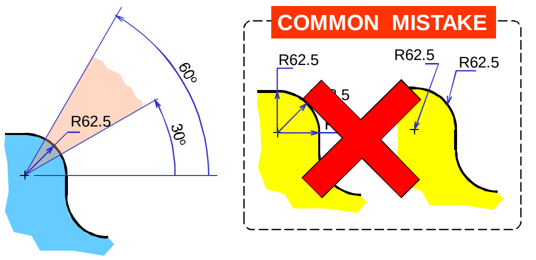
ARC

- The dimension figure and the arrowhead **should be inside** the arc, where there is sufficient space.



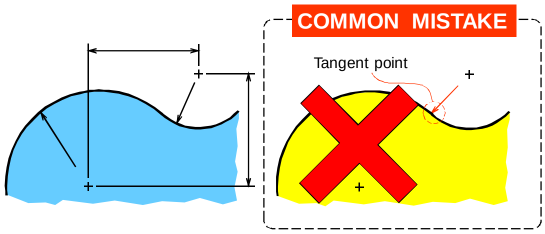
ARC

- Leader line **must** be **radial** and **inclined** with an angle between 30 ~ 60 degs to the horizontal.



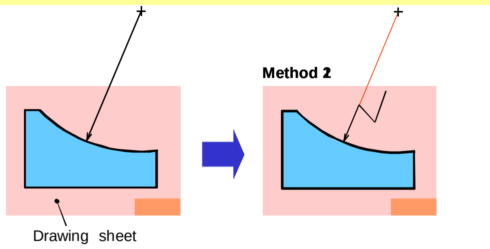
CURVE

- The curve constructed from two or more arcs, requires the dimensions of **radii** and **center's location**.



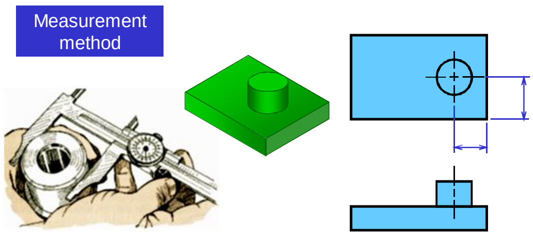
ARC

- Use the **foreshortened radial dimension line**, when arc' s center locates outside the sheet or interfere with other views.



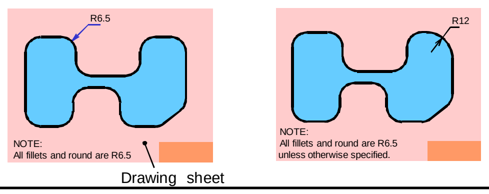
CYLINDER

- Size dimensions are **diameter** and **length**.
- Location dimension **must** be located from its center lines and **should be** given in circular view.



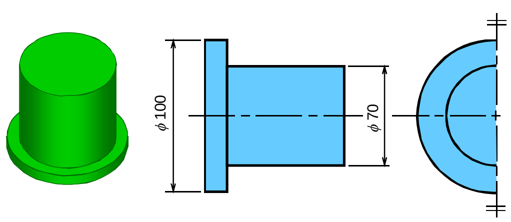
FILLETS AND ROUNDS

- Give the radius of a typical fillet only by using a ***local*** note.
- If all fillets and rounds are uniform in size, dimension may be omitted, but it is necessary to add the note " All fillets and round are Rxx. "



CYLINDER

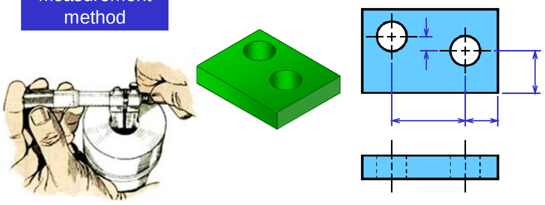
- Diameter **should be** given in a longitudinal view with the symbol "ϕ" placed before the figures.



HOLES

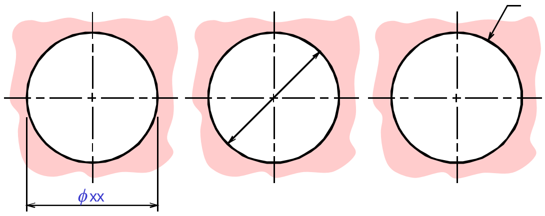
- Size dimensions are **diameter** and **depth**.
- Location dimension **must** be located from its center lines and should be given in circular view.

Measurement method



HOLES : LARGE SIZE

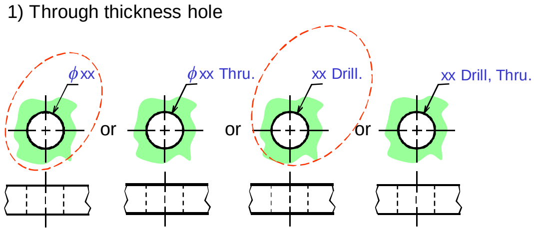
Use extension and dimension lines | Use diametral dimension line | Use leader line and note



HOLES : SMALL SIZE

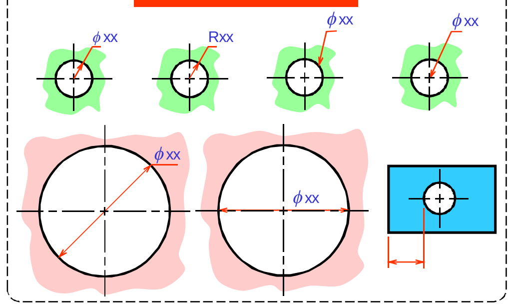
- Use leader line and local note to specify **diameter** and **hole's depth** in the circular view.

1) Through thickness hole



HOLES

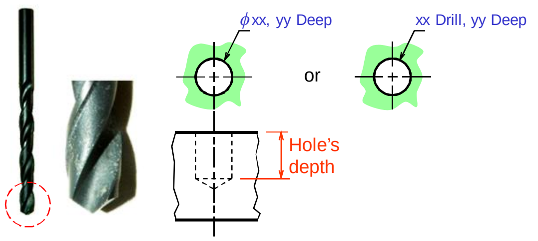
COMMON MISTAKE



HOLES : SMALL SIZE

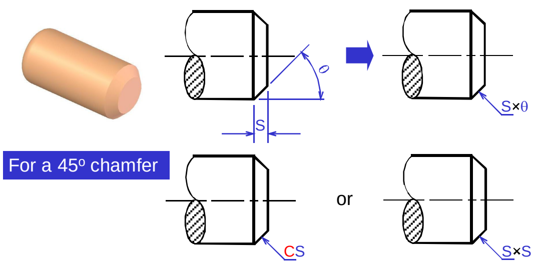
- Use leader line and local note to specify **diameter** and **hole's depth** in the circular view.

2) Blind hole



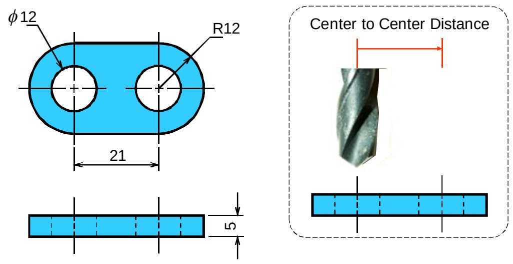
CHAMFER

- Use leader line and note to indicate **linear distance** and **angle** of the chamfer.



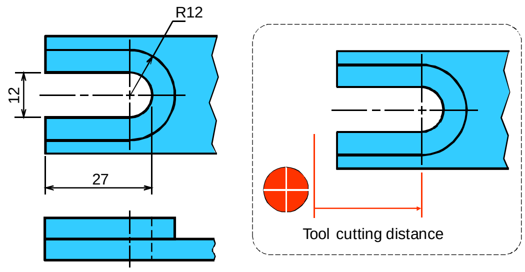
ROUNDED-END SHAPES

- Dimensioned according to the manufacturing method used.



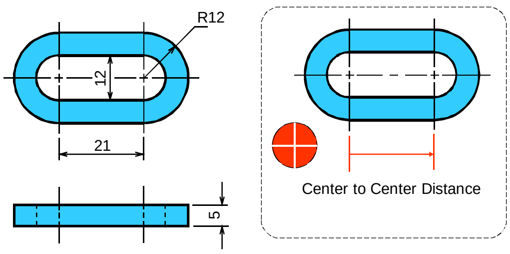
ROUNDED-END SHAPES

- Dimensioned according to the manufacturing method used.



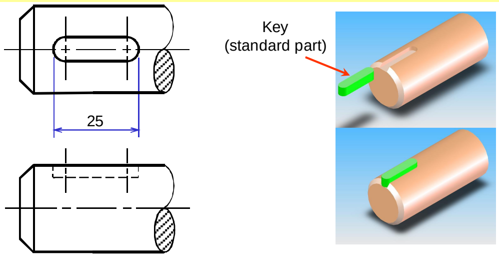
ROUNDED-END SHAPES

- Dimensioned according to the manufacturing method used.



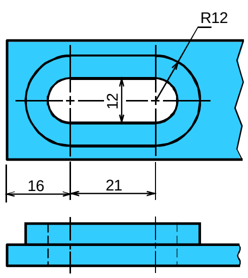
ROUNDED-END SHAPES

- Dimensioned according to the **standard sizes** of another part to be assembled or manufacturing method used.



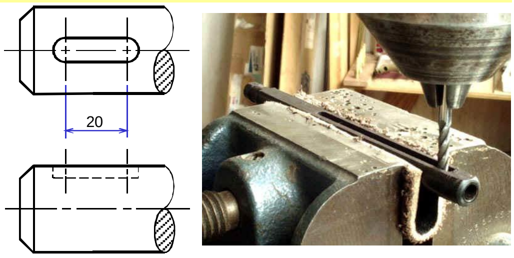
ROUNDED-END SHAPES

- Dimensioned according to the manufacturing method used.



ROUNDED-END SHAPES

- Dimensioned according to the **standard sizes** of another part to be assembled or manufacturing method used.

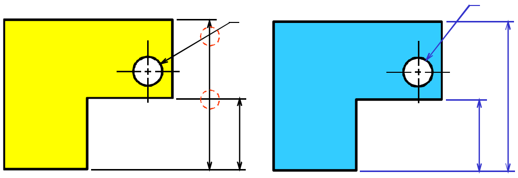


Placement of Dimensions

RECOMMENDED PRACTICE

1. Extension lines, leader lines **should not** cross dimension lines.

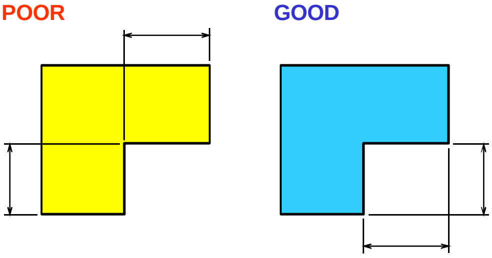
POOR GOOD



RECOMMENDED PRACTICE

2. Extension lines **should be** drawn from the nearest points to be dimensioned.

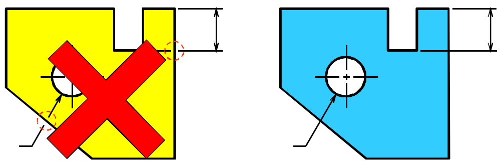
POOR GOOD



RECOMMENDED PRACTICE

3. Extension lines of internal feature ***can*** *cross visible lines* ***without*** *leaving a gap at the intersection point.*

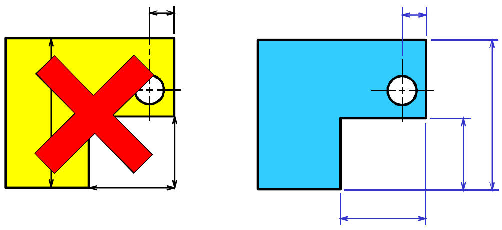
WRONG CORRECT



RECOMMENDED PRACTICE

4. ***Do not*** *use object line, center line, and dimension line as an extension lines.*

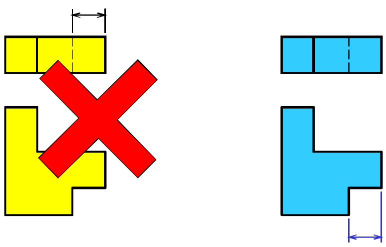
POOR GOOD



RECOMMENDED PRACTICE

5. ***Avoid*** *dimensioning hidden lines.*

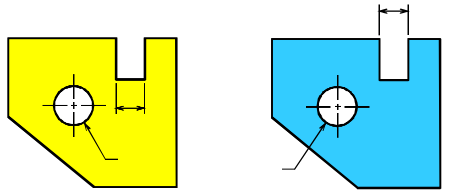
POOR GOOD



RECOMMENDED PRACTICE

6. Place dimensions **outside** the view, unless placing them inside improve the clarity.

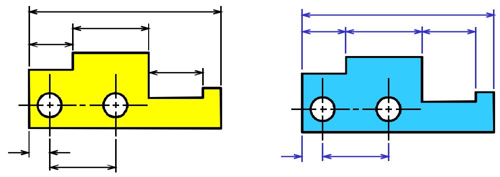
POOR GOOD



RECOMMENDED PRACTICE

8. Dimension lines should be lined up and grouped together as much as possible.

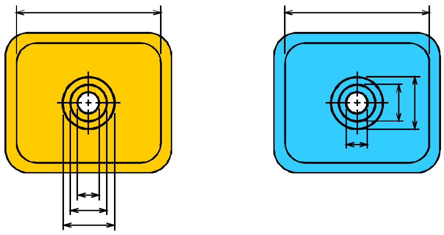
POOR GOOD



RECOMMENDED PRACTICE

6. Place dimensions **outside** the view, unless placing them inside improve the clarity.

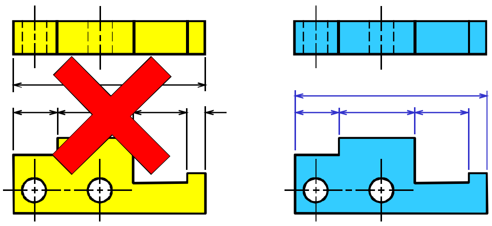
JUST OK !!! BETTER



RECOMMENDED PRACTICE

*9. **Do not** repeat a dimension.*

POOR GOOD



RECOMMENDED PRACTICE

7. Apply the dimension to the view that clearly show the shape or features of an object.

POOR GOOD